

# Nilesh Patil

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## SUMMARY

Currently building, scaling and applying generative-AI systems at **DreamStreet** - in a regulated, high-stakes, and production environment for Indian capital markets. I work on designing reliability, adherence to specs, and evaluation frameworks for LLM-based antigenic systems and deploy self-hosted LLM stacks across *GCP / AWS*. Particularly interested in AI harness design, developer productivity, and turning emerging model capabilities into reliable workflows and products.

Previously led cross-continent research and engineering teams at **Dream11** & collaboration with **Columbia University**, building production ML systems at scale and pursuing high impact research. Specifically - designing and instrumenting ML system, training ML models & transformers, supporting feature and serving infrastructure for 250M+ users. Publications in *Cell / ACL / UMAP / ECMLPKDD / ICMLA / CODS / IEEE*, and am interested in pursuing novel applications for agentic systems blending traditional machine learning problems.

[Github](#) | [Website](#) | [Medium](#) | [LinkedIn](#) | [Google Scholar](#)

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## EDUCATION

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| 2017 | M.S. in Data Science, University of Rochester                                     | Rochester, New York |
| 2013 | Bachelor of Technology in Meta. & Materials Engg., Indian Institute of Technology | Roorkee, India      |

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## RESEARCH EXPERIENCE

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| 2024 — 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Senior Principal Research Scientist, Dream Sports             | Mumbai   New York |
| <ul style="list-style-type: none"><li>Established and led an industry-academia research partnership with Columbia University, co-defining research agendas at the intersection of ML and sports analytics and supporting joint work with faculty, postdocs, and PhD students.</li><li>Defined a multi-year applied-research portfolio - formulating ~23 candidate research problems and securing funding for 10 - spanning sports robotics, LLM-based persona/behavior simulators, and agentic evaluators for personalization.</li><li>Developed deep-learning models for early churn prediction from large-scale user-product interaction time series (<i>published, ICMLA 2023</i>), and LLM-based behavioral simulation of user lifetime trajectories at population scale (250M+ users).</li><li>Built distributed recommendation, semantic similarity search over ~100M entities, and feature infrastructure for 250M+ users - work that seeded publications on personalization (<i>UMAP '25, ECML-PKDD '25</i>) and entity resolution (<i>ACL 2026</i>).</li><li>Designed real-time probabilistic system forecasting producing 50k+ concurrent forecasts for high-performance production ML applications</li></ul> |                                                               |                   |
| 2017 — 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Staff Data Scientist, Center for Vaccine Biology & Immunology | Rochester, NY     |
| <ul style="list-style-type: none"><li>Developed deep-learning pipelines for biomedical image analysis - including 3D reconstruction from hyperspectral microscopy contributing computational methods to immunology research published in <i>Cell Reports</i> and the <i>Journal of Immunology</i></li><li>Automated ultrasound Doppler-angle estimation using DNNs on ultrasound scans of (<i>EMBC 2019</i>)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                               |                   |

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## EMPLOYMENT HISTORY

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| 2026 — Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Head of AI, DreamStreet                      | Mumbai |
| <ul style="list-style-type: none"><li><i>Designed a compliance-aware orchestration framework for multi-step LLM agents in a regulated (SEBI) financial setting - incorporating constrained tool-use, guardrail/policy enforcement, and end-to-end audit logging for controllability and traceability of agent decisions.</i></li><li><i>Deployed self-hosted open source language models ( Qwen 3.5 / 3.6 27B ) using vLLM and an agent-tooling stack across on-prem, GCP, and AWS.</i></li><li>Deployed Hermes based self-serve agents for in-house productivity tasks [ domain research reviews, alerts etc ]</li></ul> |                                              |        |
| 2019 — 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Senior Principal Research Scientist, Dream11 | Mumbai |
| <ul style="list-style-type: none"><li><i>Led applied AI research for Dream Sports and built a high-performing cross-continent team of research scientists, applied scientists, and ML engineers across India and New York.</i></li><li><i>Delivered regular data, ML, and AI training sessions across Dream Sports for audiences ranging from ~10 to 200 participants.</i></li></ul>                                                                                                                                                                                                                                      |                                              |        |

- *Led the team working on distributed forecasting & recommendation systems, text similarity search for ~100M entities, and feature-store systems supporting 250M+ users*

2017 — 2019

Staff Data Scientist, David H. Smith Center for Vaccine Biology and Immunology, University of Rochester

Rochester, NY

*Built automated and self-serve ML systems for bio-imaging research, including 3D reconstruction from hyper-spectral microscopic scans.*

2014 — 2016

Data Scientist, AXA Insurance

Pune, India

*Built survival / mortality-forecasting models and scaled statistical pipelines (Spark, Python) for US Poluation ( ~300M+ entities ).*

2013 — 2014

Data Analyst, AbsolutData Research & Analytics

Gurgaon, India

*Developed a multi-stage predictive-maintenance model fusing sensor telemetry, oil-analysis, and human-labeled alerts.*

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## SELECTED PUBLICATIONS

2019 — 2026

Sr. Principal Research Scientist, Dream11

- **Structure-Guided Entity Resolution: Fine-Tuning LLMs for Robust Name Matching in Complex Linguistic Contexts**, 2026, Association for Computational Linguistics, USA. <https://doi.org/10.48550/arXiv.2605.23597>
- **Optimizing Fantasy Sports Team Selection with Deep Reinforcement Learning**, In Proceedings of CODS-COMAD '24. Association for Computing Machinery, New York, NY, USA, 284–291. <https://doi.org/10.1145/3703323.3703743>
- **Early churn prediction from large scale user-product interaction time series**. 2023 International Conference on Machine Learning and Applications (ICMLA). IEEE, 2023. <https://doi.org/10.1109/ICMLA58977.2023.00314>

2016 — 2019

Staff Data Scientist

Rochester, NY

- **Automated Ultrasound Doppler Angle Estimation Using Deep Learning**. In 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 28-31. IEEE, 2019 <https://pubmed.ncbi.nlm.nih.gov/31945837>
- **CXCL10+ peripheral activation niches couple preferred sites of Th1 entry with optimal APC encounter**, Accepted to Cell Reports <https://doi.org/10.1016/j.celrep.2021.109523>
- **CXCL10+ perivascular clusters nucleate Th1 cell tissue entry and activation in the inflamed skin**. Accepted into Journal of Immunology : <https://doi.org/10.4049/jimmunol.204.Supp.220.9>

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## RESEARCH SUPERVISION / MENTORSHIP

2024 — 2026

Sr. Principal Research Scientist, Dream Sports

- **FENCE: Fairplay Ensuring Network Chain Entity for Real-Time Multiple ID Detection at Scale In Fantasy Sports**. In Proceedings of the Third International Conference on AI-ML Systems (AIMLSystems '23). Association for Computing Machinery, New York, NY, USA, Article 26, 1–7. <https://doi.org/10.1145/3639856.3639882>
- **Maximizing Success Rate of Payment Routing using Non-stationary Bandits**. In Proceedings of the Third International Conference on AI-ML Systems (AIMLSystems '23). Association for Computing Machinery, New York, NY, USA, Article 27, 1–7. <https://doi.org/10.1145/3639856.3639883>
- **Accelerating Causal Algorithms for Industrial-scale Data: A Distributed Computing Approach with Ray Framework**. In Proceedings of the Third International Conference on AI-ML Systems (AIMLSystems '23). Association for Computing Machinery, New York, NY, USA, Article 24, 1–6. <https://doi.org/10.1145/3639856.3639880>
- **ForeCal: Random Forest-based Calibration for DNNs**. In Proceedings of the 8th International Conference on Data Science and Management of Data (12th ACM IKDD CODS and 30th COMAD) (CODS-COMAD '24). Association for Computing Machinery, New York, NY, USA, 44–51. <https://doi.org/10.1145/3703323.3703330>
- **Driving Engagement in Daily Fantasy Sports with a Scalable and Urgency-Aware Ranking Engine**. Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 40. No. 47. 2026. <https://doi.org/10.1609/aaai.v40i47.41479>
- **Enhancing Personalisation in Fantasy Sports with Graph-Based Representations**. In Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '25). Association for Computing Machinery, New York, NY, USA, 361–363. <https://doi.org/10.1145/3699682.3730974>
- **Personalized Contest Recommendation in Fantasy Sports**. In *Machine Learning and Knowledge Discovery in Databases*. Applied Data Science Track: European Conference, ECML PKDD 2025, Porto, Portugal, September 15–19, 2025, Proceedings, Part IX. Springer-Verlag, Berlin, Heidelberg, 22–35. [https://doi.org/10.1007/978-3-032-06118-8\\_2](https://doi.org/10.1007/978-3-032-06118-8_2)